Lab 4 – VibeCoding: Chat History

Name: Nishkal M R | SRN: PES1UG24AM181
Repo: SETAPESU26/54_skyscrapper_stack (Skyscraper Stack, Pygame)
AI tool: Claude (Claude Code, model Claude Sonnet 5.5)
Note: my replies are condensed; my own messages are verbatim.

Me: complete the lab4 for the srn PES1UG24AM181 and zip the file so that i can add the submission to the github repo

Claude: Found Lab_4_VibeCoding_Student_handout.pdf in the connected folder. The handout asks to fix the broken code and add the README features in the assigned repo, one commit per task, with before/after videos and the chat history. It does not name the repo, so I asked for the link. I also said the 10-second screen recordings must be made by the student.

Me: https://github.com/SETAPESU26/54_skyscrapper_stack

Claude: Cloned the repo and read the README and code (main.py, game/block.py, game/game_engine.py). Findings: (1) the deliberate bug is game_engine.py line 61, `is_successful_drop = overlap <= 0`, which is inverted: landing on the tower ends the game and a miss keeps building; (2) the README says Space restarts after Game Over but the code only handles R and click. Plan: one commit per task (1 bug fix, then Tasks 2-4).

Prompt 1 (Task 1): Fix the inverted overlap placement bug: valid overlaps should stack, complete misses should collapse the tower. Also make Space restart as the README says.

Claude: Changed the condition to `overlap > 0` and added K_SPACE to the restart check. First attempt rewrote the file with LF line endings and showed every line as changed in the diff; I redid it keeping the original CRLF endings so the commit shows only 2 changed lines.

Prompt 2 (Task 2): Add Perfect Placement: if a drop is almost flush with the tower, snap it without trimming, show a PERFECT! prompt with bonus score, and restore some width after consecutive perfects.

Claude: Added a 6 px tolerance, snap-to-top-block, a floating fading PERFECT! popup, streak-scaled bonus (5 x streak) shown in the HUD, and +8 px width restored per perfect from the 2nd in a row (capped at the 180 px base width, kept inside the arena). A normal drop resets the streak. Headless test checked: stacking, snap, bonus, restore, cap, reset, collapse, Space restart.

Prompt 3 (Task 3): Add falling, rotating debris for the sliced-off part of a block.

Claude: Added game/debris.py (gravity, spin, fade, removal off-screen). Each trimmed overhang becomes a piece; a complete miss makes the whole block tumble. Debris moves with the camera scroll. First test failed only because the piece had already fallen off-screen in 30 frames, so I shortened the test; the code was fine.

Prompt 4 (Task 4): Make the background shift through atmospheric colour gradients as the tower gets taller.

Claude: Added a vertical gradient interpolated between day, sunset, dusk, night and space keyframes. The driving height eases toward the real score so the shift is smooth; stars fade in at night. Rendered frames at heights 0/8/16/26/40 to check the colours. A bot then played 262 games without a crash.